



Joint low-rank prior and difference of Gaussian filter for magnetic resonance image denoising

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Abstract

The low-rank matrix approximation (LRMA) is an efficient image denoising method to reduce additive Gaussian noise. However, the existing low-rank matrix approximation does not perform well in terms of Rician noise removal for magnetic resonance imaging (MRI). To this end, we propose a novel MR image denoising approach based on the extended difference of Gaussian (DoG) filter and nonlocal low-rank regularization. In the proposed method, a novel nonlocal self-similarity evaluation with the tight frame is exploited to improve the patch matching. To remove the Rician noise and preserve the edge details, the extended DoG filter is exploited to the nonlocal low-rank regularization model. The experimental results demonstrate that the proposed method can preserve more edge and fine structures while removing noise in MR image as compared with some of the existing methods.

Keywords Difference of Gaussian (DoG) filter · Low-rank matrix approximation (LRMA) · MR image denoising · Nonlocal self-similarity · Singular value thresholding

1 Introduction

The denoising of MR image is proven to be a useful medical image processing technique for clinical investigation and disease diagnosis [21, 28, 37]. It is widely used for pre-processing, such as common MRI [39] and dynamic cardiac MR images [24]. MR images usually require more texture feature and edge details, compared with other types of images. These texture features should be better preserved as they provide a better reference of imaging technology to diagnose [26, 40]. In general, noise in magnitude MRI is assumed as the following Rician distribution [25]. How to remove the Rician noise from a noisy MR image is a challenging task.

Numerous methods for the denoising of MR images were developed over the decades. In [18], a variational model based on total variation-based minimization is studied on

Rician noise denoising. In [6], a Rician noise removal model with dictionary learning is proposed to prevent the block artifact. Unfortunately, it wiped some fine structures and also deteriorated the image quality. In [36], a non-local means (NLM) filter is proposed to remove the Rician noise, and causes the edge oscillations among sharp edges of an image. To overcome this issue, the variance-stabilization approach for Rician noise removal is proposed in the arena of MR image denoising [16]. Recently, the low-rank matrix approximation (LRMA) with nuclear norm minimization (NNM) has been successfully developed to remove the Gaussian noise [5, 8]. It is found that the matrix consists of similar patches of image; it has a low-rank structure property. Therefore, the nonlocal self-similarity (NSS) prior of images is utilized to construct the nonlocal LRMA model, which can obtain significant denoising effects [17, 44, 49]. Despite good theoretical guarantee by the classic technique of singular value thresholding (SVT), the NNM is not accurate enough to approximate the matrix rank [20]. The reason is that the solution depends on thresholding values. To address this problem, Gu et al. [20] propose a weighted NNM (WNNM) to keep the large singular values and achieve competitive results. In [22], the authors propose a novel image denoising method by exploiting the matrix rank minimization and the patch-based NSS structure. Unfortunately, applying these techniques directly to MR image denoising

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may result in obvious noise residual and edge blur, even edge structure loss. The reason is that the nuclear norm type models are often designed to remove Gaussian noise, but noise in the magnitude MR image is usually modeled as Rician distribution. Subsequently, this technique is employed to the WNNM model to further improve the denoising performance, termed WNNMD-N [47]. However, it still loses some high-frequency components of the image, like edges and fine structures, which are hard to discriminate in the anatomical structures of MR images.

To better interpret this problem, we show the example of MR image denoising in Fig. 1. It is observed that some of edge textures are blurred in Fig. 1b. Obviously, the denoising result of WNNM preserves the edge textures in detail with difficulty. Recently, some deep learning-based methods are used for the MR image denoising that can achieve impressive results with a high computational cost [27]. However, the methods usually require large training samples and the denoising performance relies on the training samples.

To relieve these concerns, a novel MR image denoising approach is proposed that aims to investigate and improve the application of LRMA to the denoising of MR data. Inspired by the study of the difference of Gaussian (DOG) filter, it has strong edge detection abilities [2, 46]. On this account, the denoising approach presented in this paper is to combine nonlocal low-rank regularization model with the extended DoG filter. To construct the nonlocal low-rank matrix, a similarity evaluation scheme based on the tight frame is exploited on a noisy patch to avoid the effect of the noise in the patch matching. Since the result obtained by WNNM-based type denoising may lead to the high-frequency components' loss of the image (e.g., edges and fine structures), the extended DoG filter is exploited to retrieve the high-frequency components. To solve the nonlocal low-rank model, the equivalent model is deduced. The experimental results illustrate that the accuracy of the proposed method is better than those yielded by some of the

existing MRI denoising methods. The major contributions of this paper are summarized as follows:

- A novel nonlocal self-similarity evaluation scheme based on the tight frame is proposed for the patch matching. Different from previous nonlocal self-similarity strategy, the proposed scheme considers the effect of the noises on patch matching.
- The DoG filter is exploited to retrieve the loss of edge details of image. It avoids removal of the edge details, when the WNNM model for denoising method is executed.
- Instead of solving the nonconvex low-rank model, the equivalent model is proposed, which is reformulated into a tractable form to implement the low-rank approximation.
- Finally, an iterative algorithm is established to implement MR image denoising.

The rest of this paper is organized as follows. Section 2 reviews the previous work. Section 3 details the proposed method and enhances the resolution of the MR image using the DoG filter. Extensive experiments on several databases are conducted in Section 4. Finally, the conclusion is given in Section 5.

2 Background and related works

2.1 The low-rank matrix approximation

Typically, the image denoising aims to estimate a latent noiseless MR data $\mathbf{X} = \text{Re}(\mathbf{X}) + j\text{Im}(\mathbf{X}) \in \mathbb{C}^{M \times N}$ from its noisy observation:

$$\mathbf{Y} = \mathbf{X} + \mathbf{E}(0, \sigma^2) = \mathbf{X} + \sigma^2 \mathbf{E}(0, 1), \quad (1)$$

where $\mathbf{Y} \in \mathbb{C}^{M \times N}$ is the noisy image and $\mathbf{E}(0, 1) = \text{Re}(\mathbf{E}(0, 1)) + j\text{Im}(\mathbf{E}(0, 1)) \in \mathbb{C}^{M \times N}$ denotes the noise,

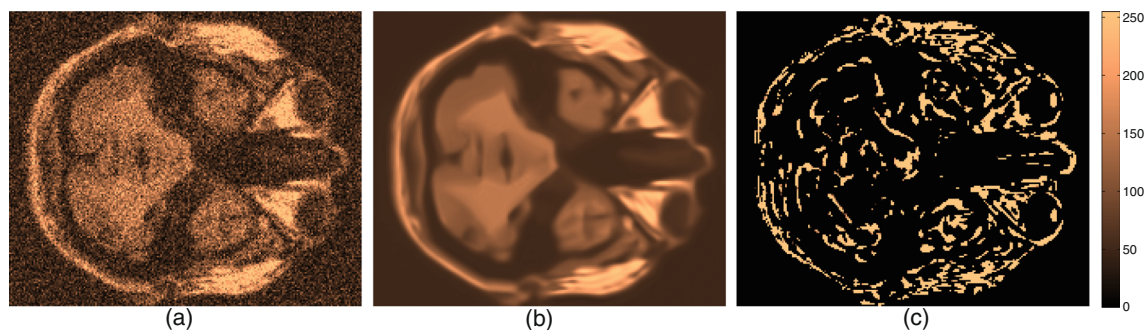


Fig. 1 MR image denoising. **a** Noisy MR image ($\sigma^2 = 30$). **b** Denoised MR image by WNNM. **c** The lost high-frequency components of the image

$\text{Re}(E(0, 1))$, $\text{Im}(E(0, 1))$ are independent and identically distributed (i.i.d.) entries that obey the standard normal distribution, and σ^2 is the noise variance.

For the clinical diagnosis, the complex MR data is generally converted into its magnitude and phase data because these are more involved with the vision of interest from the anatomy and physiology [19]. Thus, the magnitude representation of the noisy image is expressed as:

$$\mathbf{Z} = |\mathbf{Y}| = \sqrt{\text{Re}(\mathbf{Y})^2 + \text{Im}(\mathbf{Y})^2}, \quad (2)$$

where $|\cdot|$ is to calculate the magnitude of a matrix.

How to obtain a noise-free image $\mathbf{\Gamma}$ (the magnitude of MR data $\mathbf{X}$) is non-trivial. Many image prior models have been developed in literature to remove the noise from the natural image. A well-known method is the low-rank model, which is widely used in image processing and computer vision [23][3][15]. The model is formulated as:

$$\min_{\mathbf{\Gamma}} \frac{1}{2} \|\mathbf{Z} - \mathbf{\Gamma}\|_F^2 + \lambda \text{rank}(\mathbf{\Gamma}), \quad (3)$$

where $\text{rank}(\cdot)$ denotes the rank of a matrix.

Since the model (3) is a nonconvex NP-hard problem, it is generally relaxed by substitutively minimizing the NNM. Recently, a weighted NNM method is proposed, leading to very impressive results [20]. The rank function is defined as $\|\mathbf{\Gamma}\|_{*,w} \triangleq \sum_{i=1}^r w_i \sigma_i(\mathbf{\Gamma})$, $\sigma_i(\mathbf{\Gamma})$ is the i -th singular value of $\mathbf{\Gamma}$ and w_i is the i -th weight. To solve this model, the SVT operator $\mathbb{S}_{\lambda,w}(\cdot)$ is usually used by carrying out Singular Value Decomposition (SVD) on the matrix [20]. The SVD of $\mathbf{Z}$ is $\mathbf{U}\mathbf{\Sigma}_Z\mathbf{V}^H$ and $\sigma_i(\mathbf{Z})$ is the i -th singular value of the $\mathbf{\Sigma}_Z$. The SVT on the matrix $\mathbf{Z}$ is formulated as

$$\mathbb{S}_{\lambda,w}(\mathbf{Z}) \triangleq \mathbf{U}\mathbb{S}_{\lambda,w}(\mathbf{\Sigma}_Z)\mathbf{V}^T,$$

where the i -th diagonal element of $\mathbb{S}_{\lambda,w}(\mathbf{\Sigma}_Z)$ is $S_{\lambda,w}(\mathbf{\Sigma}_Z)_i$ and

$$S_{\lambda,w}(\mathbf{\Sigma}_Z)_i = \text{sgn}(\sigma_i(\mathbf{Z})) \cdot \max(\sigma_i(\mathbf{Z}) - \lambda w_i, 0).$$

It has been proven that the model (3) combined with the patch-based NSS priori of the image may produce significant denoising effects, such as RM [22] and WSNMD-3D [50].

2.2 Nonlocal self-similarity estimation

As we know, an image generally contains nonlocal self-similarity, which can be developed to improve the denoising performance [48]. Therefore, the NSS of the image is exploited as the prior knowledge to low-rank model [20, 47].

First, the noisy image $\mathbf{Z}$ is divided into a series of overlapping patches, and vectorizing these patches into a one-dimensional vector denoted by $\{\hat{\mathbf{p}}_l\}_{l=1,2,\dots,L}$, where L is the number of the patches. For each patch $\hat{\mathbf{p}}_l$, its similarity patches are searched and vectorizing each similarity patch into vector denoted as $\{\hat{\mathbf{p}}_c\}_{c=1,2,\dots,C}$, where C is the number of similarity patches. To reduce the computational cost, we build search windows of size $h \times h$ to search its similarity patches. The details can be seen in [42].

The Euclidean Distance Measure (EDM) is used to determine the similarity of two patches. A smaller EDM value implies high similarity in the structure of two patches. Therefore, the traditional calculation of similarity between an exemplar patch $\hat{\mathbf{p}}_l$ of size $a \times a$ and a candidate one $\hat{\mathbf{p}}_c$ can be formulated as:

$$\text{EDM} = \|\hat{\mathbf{p}}_c - \hat{\mathbf{p}}_l\|_2^2, \quad (4)$$

where $\hat{\mathbf{p}}_c$, ($c = 1, \dots, C$), are the candidate patches with respect to the exemplar patch $\hat{\mathbf{p}}_l$.

The drawback of this method is its patch matching from the noisy image, that gives rise to inaccuracy in estimation of true patch matching. Hence, the EDM-based similarity criterion may lead to patch mismatching because comparison of noise is meaningless.

To suppress the effect of noise on the patch matching, the patch preprocessing is proposed, which is a more effective than the conventional one. The sparse representation of the tight frame is also used for the patch matching, since the tight frame representation has a remarkable sparse approximation and low complexity [4][41]. Given a tight frame $\mathbf{D}_l$, the sparse representation of the image patch $\mathbf{p}_l$ is $\mathbf{s}_l = \mathbf{D}_l^T \mathbf{p}_l$, where $\mathbf{s}_l$ is the sparse coefficient.

By exploiting the property of the tight frame, the image-free patch of the EDM is used to obtain its similar patches, that is

$$\text{EDM} = \|\mathbf{p}_c - \mathbf{p}_l\|_2^2 \stackrel{(a)}{=} \|\mathbf{s}_c - \mathbf{s}_l\|_2^2 \quad (5)$$

where $\mathbf{p}_c$, $\mathbf{p}_l$ are the noise-free patches corresponding the noisy patches $\hat{\mathbf{p}}_c$ and $\hat{\mathbf{p}}_l$, respectively. (a) is from the fact that the tight frame property $\mathbf{D}_l^T \mathbf{D}_l = \mathbf{I}$ and

$$\|\mathbf{p}_c - \mathbf{p}_l\|_2^2 = (\mathbf{p}_c - \mathbf{p}_l)^T \mathbf{D}_l^T \mathbf{D}_l (\mathbf{p}_c - \mathbf{p}_l) = \|\mathbf{s}_c - \mathbf{s}_l\|_2^2$$

As shown in [38], the noise-free images and noise are uncorrelated. According to Eq. (1), for each element $\hat{s}_l[t]$ of noisy patch $\hat{\mathbf{s}}_l$, we have

$$E\{\hat{s}_l^2[t]\} = s_l^2[t] + 2\sigma^2, \quad E\{\hat{s}_c^2[t]\} = s_c^2[t] + 2\sigma^2, \quad (6)$$

where σ^2 is the noise variance.

Thus, the patch preprocessing can be expressed as

$$s_l[t] = \sqrt{E\{\hat{s}_l^2[t]\} - 2\sigma^2}, \quad s_c[t] = \sqrt{E\{\hat{s}_c^2[t]\} - 2\sigma^2}, \quad (7)$$

where $p_l[t]$, ($t \in \{1, 2, \dots, a^2\}$), is the corresponding element of the p_l in the patch data. By Eq. (7), the similarity patches can be obtained with Eq. (5).

For the formula (7), the noise variance σ^2 needs to be estimated first. In practical situations, the characteristics of Rician noise depend on signal-to-noise ratio (SNR) [30]. In low-SNR values, noise in the magnitude MR image can be modeled by a Rayleigh distribution. Otherwise, the noise distribution in high-SNR values can be well approximated using a Gaussian distribution [29]. To obtain the spatially varying noise map σ , the robust noise standard deviation should be estimated in different SNR values as follows:

$$\sigma = \frac{1}{\sqrt{\xi(\theta)}} \sigma_0 \quad (8)$$

where θ is the estimated SNR value and the correction factor $\xi(\theta)$ is defined as [30]:

$$\xi(\theta) = 2 + \theta^2 - \frac{\pi}{8} \exp\left(-\frac{\theta^2}{2}\right) \times \left((2 + \theta^2) I_0\left(\frac{\theta^2}{4}\right) + \theta^2 I_1\left(\frac{\theta^2}{4}\right)\right) \quad (9)$$

and

$$\theta = \sqrt{\max\left(\frac{E(\hat{s}_l^2)}{\sigma_0^2} - 2, 0\right)},$$

σ_0 is the first result of estimation, which is estimated as [11]

$$\sigma_0 = \frac{\text{Median}(|(\hat{s}_l)_j|)}{0.6745}, \quad (\hat{s}_l)_j \in \text{subband } HH_1 \quad (10)$$

The above similarity metric will be introduced to obtain more accurate patch matching in MR image denoising.

3 Enhanced edge-preserving for MR image denoising

In this section, we present a general denoising framework based on nonlocal low-rank regularization model and the extended DoG filter. Firstly, the nonlocal self-similarity structure of image is exploited to get a series of patches. Then, using the similarity criterion in Eq. (5), the similar image patches are formed by the patch preprocessing.

Inspired by the study [46], the extended DOG filter is exploited to capture the edge details. Thus, the lost high-frequency components are retrieved by the extended DoG filter, when the WNNM denoising method is executed to remove the Rician noise. Finally, a relaxation parameter is designed to suppress the effects of block overlap.

The algorithm consists of four stages:

3.1 Block matching

Let $\hat{p}_l = G_l \mathbf{Z}$ be a vectorized format of the l -th image patch, where G_l is the matrix to extract the patch $\hat{p}_l$ from $\mathbf{Z}$ at location l (see [48] for details). Thus, the noisy MR image $\mathbf{Z}$ is divided into L overlapping patches denoted by $\{\hat{p}_l\}_{l=1,2,\dots,L}$, where $\hat{p}_l = [\hat{p}_l[1], \hat{p}_l[2], \dots, \hat{p}_l[a^2]]^T$. For each exemplar patch $\hat{p}_l$, its C nearest patches are obtained by the novel nonlocal self-similarity evaluation scheme used in Eq. (5), and denoted by $\{\hat{p}_{l_1}, \hat{p}_{l_2}, \dots, \hat{p}_{l_C}\}$. By the patch preprocessing, the group matrix $\mathbf{Z}_l \in \mathbb{R}^{a^2 \times C}$ is constructed as

$$\mathbf{Z}_l = [\mathbf{p}_l, \mathbf{p}_{l_1}, \mathbf{p}_{l_2}, \dots, \mathbf{p}_{l_C}] \quad (11)$$

to denote the l -th group of the similar patches.

3.2 The low-rank model for denoising

In this stage, the basic task is to estimate the denoised matrix $\mathbf{S}_l$ from its noisy version $\mathbf{Z}_l$ as accurate as possible. The denoised group matrix $\mathbf{S}_l$ consists of similar patches that lead to high relativity with column structure, which means that $\mathbf{S}_l$ is a low-rank matrix. Thus, the estimate of $\mathbf{S}_l$ can be obtained by minimizing the following optimization problem:

$$\min_{\mathbf{S}_l} \frac{1}{2} \|\mathbf{Z}_l - \mathbf{S}_l\|_F^2 + \lambda \|\mathbf{S}_l\|_{*,w} \quad (12)$$

where $\|\mathbf{S}_l\|_{*,w} = \sum_{i=1}^r w_i \sigma_i(\mathbf{S}_l)$ and $r = \min\{a^2, C\}$.

Since Eq. (12) is nonconvex, the SVT is used in solving, which makes this to fall into the suboptimal solution. To solve the optimization problem (12), the equivalent model is given to achieve the closed form solution. That is

$$\min_{\mathbf{S}_l} \frac{1}{2} \|\mathbf{W}_l \mathbf{Z}_l - \mathbf{W}_l \mathbf{S}_l\|_F^2 + \lambda \|\mathbf{W}_l \mathbf{S}_l\|_* \quad (13)$$

where the weight matrix $\mathbf{W}_l = \text{diag}\{w_{l_1}, w_{l_2}, \dots, w_{l_r}\}$ is a diagonal matrix and w_{l_i} is i -th weight value. Here, we

define Φ_l as $\Phi_l \triangleq U_l^T L_{S_l} V_l$, where L_{S_l} denotes the sparse representation of S_l in the tight frame domain (i.e., $L_{S_l} = D_l S_l$). Similarly, the L_{Z_l} is the tight frame representation of Z_l and the corresponding SVD is $L_{Z_l} = U_l \Sigma_{Z_l} V_l^T$. The proof is provided in [Appendix](#).

Obviously, the optimization problem (13) can be rewritten as

$$\min_{\sigma_1(\Phi_l), \dots, \sigma_r(\Phi_l)} \frac{1}{2} \sum_{i=1}^r (w_{l_i} (\sigma_i(Z_l) - \sigma_i(\Phi_l)))^2 + \lambda \sum_{i=1}^r w_{l_i} \sigma_i(\Phi_l) \quad (14)$$

This means that the minimizations with respect to $\sigma_1(\Phi_l), \dots, \sigma_r(\Phi_l)$ can be separated as

$$\min_{\sigma_i(\Phi_l)} \frac{1}{2} (w_{l_i} (\sigma_i(Z_l) - \sigma_i(\Phi_l)))^2 + \lambda w_{l_i} \sigma_i(\Phi_l) \quad (15)$$

From derivative analysis, the closed form solution of Eq. (15) is obtained by

$$\sigma_i(\Phi_l) = \max \left\{ \sigma_i(Z_l) - \frac{\lambda}{w_{l_i}}, 0 \right\} \quad (16)$$

where $w_{l_i} = \frac{\eta_i(Z_l)}{2\sqrt{n}\sigma^2}$ is the weight value with the same in [\[49\]](#).

It follows that the solution of Eq. (13) can be given by $\Sigma_{\Phi_l} = \text{diag}\{\sigma_1(\Phi_l), \dots, \sigma_r(\Phi_l)\}$. By the definition of Φ_l , the S_l can be obtained by

$$S_l = D_l^T U_l \text{diag}\{\sigma_1(\Phi_l), \dots, \sigma_r(\Phi_l)\} V_l^H. \quad (17)$$

where D_l denotes the sparse representation of tight frame.

3.3 Enhanced edge-preserving method via the DoG filter

As shown in “Introduction,” the denoising performance of NNM type model is affected in Rician noise that may cause higher frequency component loss in image. According to the solution (17), we define $R_l = Z_l - S_l$ as the residuals, which consists of the high-frequency components H_l and the noise $N_l(0, \sigma^2)$. That is

$$R_l^{[i]} = N_l(0, \sigma^2)^{[i]} + H_l^{[i]}, \quad i \in \{1, 2, \dots, a^2 C\} \quad (18)$$

where $R_l^{[i]}$, $H_l^{[i]}$ are the i -th element of the R_l and H_l , respectively. For each element $N_l(0, \sigma^2)^{[i]}$ of the noise $N_l(0, \sigma^2)$, it can be expressed as $N_l(0, \sigma^2)^{[i]} = \sigma^2 N_l(0, 1)^{[i]}$.

To preserve edges and fine structures, the extended DoG filter is introduced to retrieve the the higher frequency components from the residuals R_l . The extended DoG filter with spread parameters ϱ_1, ϱ_2 is defined as [\[46\]](#)

$$DoG(R_l^{[i]}) = \frac{1}{\sqrt{2\pi}} \left(G_{\varrho_1}(R_l^{[i]}) - \gamma G_{\varrho_2}(R_l^{[i]}) \right) \quad (19)$$

where $G_{\varrho_j}(R_l^{[i]}) = \frac{1}{\varrho_j} \exp\left(-\frac{(R_l^{[i]})^2}{2\varrho_j^2}\right)$, $j = 1, 2$ and γ is a scale factor that is empirically set as $\gamma = \frac{\varrho_2}{\varrho_1}$ in experiments.

However, the estimated result of the extended DoG filter still involves the inaccurate pseudo high-frequency components, which are generally considered as the high-frequency parts of noise. Hence, the effect of pseudo high-frequency components should be considered in the high-frequency component extraction based on the extended DoG filter.

To address the pseudo high-frequency components, the logarithm is used to separate low frequency σ^2 and higher frequency $N_l(0, 1)^{[i]}$ in noise;

$$\log N_l(0, \sigma^2)^{[i]} = \underbrace{\log \sigma^2}_{\text{low-frequency}} + \underbrace{\log N_l(0, 1)^{[i]}}_{\text{pseudo high-frequency}} \quad (20)$$

We apply the DoG filter to the noise $N_l(0, \sigma^2)^{[i]}$, and thus the low frequency of noise will be removed. Hence, the Eq. (20) can be reformulated as

$$\log N_l(0, \sigma^2)^{[i]} = \log \sigma^2 + \log (DoG(N_l(0, \sigma^2)^{[i]})). \quad (21)$$

To solve Eq. (21), the $DoG(N_l(0, \sigma^2)^{[i]})$ can be rewritten as

$$\begin{aligned} DoG(N_l(0, \sigma^2)^{[i]}) &= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{\varrho_1} \exp\left(-\frac{(N_l(0, \sigma^2)^{[i]})^2}{2\varrho_1^2}\right) - \frac{\gamma}{\varrho_2} \exp\left(-\frac{(N_l(0, \sigma^2)^{[i]})^2}{2\varrho_2^2}\right) \right) \\ &= \frac{1}{\sqrt{2\pi}\varrho_1} \exp\left(-\frac{(N_l(0, \sigma^2)^{[i]})^2}{2\varrho_1^2}\right) \left(1 - \exp\left(-\frac{(N_l(0, \sigma^2)^{[i]})^2}{2\hat{\varrho}^2}\right) \right) \end{aligned} \quad (22)$$

where $\hat{\varrho}^2 = \frac{\varrho_1^2 \varrho_2^2}{\varrho_2^2 - \varrho_1^2}$. By first-order Taylor expansion of $\exp\left(-\frac{(N_l(0, \sigma^2)^{[i]})^2}{2\hat{\varrho}^2}\right)$, we can obtain:

$$\begin{aligned} DoG(N_l(0, \sigma^2)^{[i]}) &= \frac{1}{\sqrt{2\pi}\varrho_1} \exp\left(-\frac{(N_l(0, \sigma^2)^{[i]})^2}{2\varrho_1^2}\right) \left[1 - \left(1 - \frac{(N_l(0, \sigma^2)^{[i]})^2}{2\hat{\varrho}^2} + \dots \right) \right] \\ &\approx \frac{(N_l(0, \sigma^2)^{[i]})^2}{2\sqrt{2\pi}\varrho_1\hat{\varrho}^2} \exp\left(-\frac{(N_l(0, \sigma^2)^{[i]})^2}{2\varrho_1^2}\right) \end{aligned} \quad (23)$$

From Eqs. (18) and (23), Eq. (21) can be written as

$$\log(\mathbf{R}_l^{[i]} - \mathbf{H}_l^{[i]}) = \log \sigma^2 + \log \left(\frac{(\mathbf{R}_l^{[i]} - \mathbf{H}_l^{[i]})^2}{2\sqrt{2\pi}\varrho_1\hat{\varrho}^2} \right) - \left(\frac{(\mathbf{R}_l^{[i]} + \mathbf{H}_l^{[i]})^2}{2\varrho_1^2} \right). \quad (24)$$

Thus, we set the derivative of Eq. (24) with respect to $\mathbf{H}_l^{[1]}, \dots, \mathbf{H}_l^{[a^2C]}$ equal to 0. We derive

$$\frac{1}{\mathbf{R}_l^{[i]} - \mathbf{H}_l^{[i]}} = \frac{2}{\mathbf{R}_l^{[i]} - \mathbf{H}_l^{[i]}} - \frac{\mathbf{R}_l^{[i]} + \mathbf{H}_l^{[i]}}{\varrho_1^2}. \quad (25)$$

After simple derivations, the optimal solution of Eq. (25) is given by

$$\begin{aligned} (\mathbf{H}_l)_1^{[i]} &= \mathbf{R}_l^{[i]} + \varrho_1, \\ (\mathbf{H}_l)_2^{[i]} &= \mathbf{R}_l^{[i]} - \varrho_1. \end{aligned} \quad (26)$$

Since $0 < \mathbf{H}_l^{[i]} < \mathbf{R}_l^{[i]}$, $(\mathbf{H}_l)_1$ could be eliminated. Hence, the solution to Eq. (24) is given by $\mathbf{H}_l^{[i]} = (\mathbf{H}_l)_1^{[i]}$.

Then, the high-frequency components $\mathbf{H}_l$ are added back to the $\mathbf{S}_l$, namely

$$\hat{\mathbf{Q}}_l = \mathbf{S}_l + \tau_l \mathbf{H}_l. \quad (27)$$

where τ_l is the relaxation parameter, which is designed on the group matrix by the Gaussian kernel function to suppress the effects of patch overlap.

The corresponding relaxation parameter is given by

$$\tau_l = \frac{1}{\xi_l} \exp \left(- \left(\frac{\|\mathbf{H}_l * \text{DoG}(\mathbf{H}_l) - \mathbf{H}_l\|_2^2}{v} \right) \right) \quad (28)$$

where v is a pre-determined scalar and ξ_l is the normalization factor [14].

After obtaining the estimate of the denoised patches, the final MR image patch is calculated by averaging all the patches with its similar patches. Thus, the denoised patch $\hat{\mathbf{F}}_l$ can be obtained by the average

$$\hat{\mathbf{x}}_l = \frac{1}{C+1} \sum_{t=1}^{C+1} \hat{q}_{lc}. \quad (29)$$

where $\hat{\mathbf{x}}_l$ is the denoised patch $\hat{\mathbf{F}}_l$ by vectorization, $\hat{q}_{lc}$ is the c -th column of denoised matrix $\hat{\mathbf{Q}}_l$.

3.4 Patches aggregating

In the last step, all of the denoised patches will be put back to their original positions. By the relationship $\hat{\mathbf{F}}_l = G_l \mathbf{\Gamma}$, the estimate of whole image is constructed by the least-square solution [48][33]:

$$\mathbf{\Gamma} = \left(\sum_{l=1}^L G_l^T G_l \right)^{-1} \sum_{l=1}^L (G_l^T \hat{\mathbf{F}}_l). \quad (30)$$

Inspired by [13][43], the back projection strategy is generally developed to further suppress the noise residuals, which can enhance the denoising performance effectively. The main idea of back projection is adding some filtered noise to generate a new noisy image in each iteration. That is

$$\mathbf{Z}^{(k+1)} = \mathbf{\Gamma}^{(k)} + \delta(\mathbf{Z} - \mathbf{\Gamma}^{(k)}) \quad (31)$$

where δ is the tuning parameter subject to the noise variance σ^2 [32].

So far, all issues have been solved. In fact, all issues are designed as each separated sub-problem, which enables easier analysis of the whole denoising procedures. To summarize, our algorithm is listed as follows:

Algorithm 1 The DoG-WNNM Algorithm.

- 1: **Initialization:** Set $\mathbf{Z}, \mathbf{\Gamma}^{(0)} = \mathbf{Z}, \mathbf{X}^{(0)} = \mathbf{0}, k = 1, \varrho_1, \varrho_2, \delta \in (0, 1)$;
 - 2: **While:** the stopping criterion is not met **do**
 - 3: Update the $\mathbf{Z}^{(k)} = \mathbf{\Gamma}^{(k-1)} + \delta(\mathbf{Z} - \mathbf{\Gamma}^{(k-1)})$
 - 4: **for** each patch p_l in $\mathbf{Z}^{(k)}$ **do**
 - 5: Update the $\mathbf{Z}_l^{(k)}$ by computing (11);
 - 6: Update the $\Phi_l^{(k)}$ by computing (13);
 - 7: Update the $\mathbf{S}_l^{(k)}$ by computing (17);
 - 8: Update the high-frequency components $\mathbf{H}_l^{(k)}$ by computing (26);
 - 9: Estimate the $\hat{\mathbf{Q}}_l^{(k+1)}$ by computing (27);
 - 10: **end for**
 - 11: The denoised MR image $\mathbf{\Gamma}^{(k+1)}$ is formed by Eq. 30;
 - 12: **end while**
 - 13: **Output** $\hat{\mathbf{F}}$
-

4 Experimental results and discussion

In this section, we validate the efficacy of the proposed DoG-WNNM in the application of MR image denoising.

To obtain a realistic human brain phantom, the “BrainWeb: Simulated Brain Database” is adopted, available at <http://brainweb.bic.mni.mcgill.ca/brainweb/> [10]. Noise-free T1-weighted data (T1w) and T2-weighted data (T2w) data are used in the experiments. We compare the proposed method with several recent denoising methods, including NLM [36], WNNM [20], and WNNMD-N [47]. The proposed method is implemented in MATLAB. The parameters of the proposed method are set as follows: image size 181×217 ; patch size 6×6 ; the search window of size $h = 45$; the scale factor γ empirically is set as 1.6, $\varrho_1 = 0.001$ and $\varrho_2 = 0.0016$. As reported previously, Gaussian additive noise can be made only in the case of high SNR values, otherwise the Rician distribution has to be considered. According to [12, 31, 34], the noisy Brainweb-simulated dataset is obtained by modeling as follows:

$$y = \sqrt{(x + e_1)^2 + e_2^2}$$

where x is the ground truth and e_1, e_2 represent independent Gaussian white noise.

Three objective criteria, the peak signal-to-noise ratio (PSNR), the relative l_2 -norm error (RLNE), and quality index based on local variance (QILV) [45], are adopted to provide quantitative evaluation. Given a true MR image x , the PSNR index is defined by [7]:

$$\text{PSNR} = 10 \log_{10} \left(\frac{1}{\|\hat{x} - x\|_2^2} \right)$$

and the RLNE index is calculated by:

$$\text{RLNE} = \frac{\|\hat{x} - x\|_2}{\|x\|_2},$$

where $\hat{x}$ is the denoised image. A smaller RLNE value implies higher consistencies between denoised and original images.

The QILV index is also used for the quantitative evaluation. Compared to PSNR and RLNE, the QILV index can comprehensively reflect the performance of the denoising results. The QILV can be defined as

$$\text{QILV} = \frac{2\mu_x\mu_{\hat{x}}}{\mu_x^2 + \mu_{\hat{x}}^2} \frac{2\sigma_x\sigma_{\hat{x}}}{\sigma_x^2 + \sigma_{\hat{x}}^2} \frac{\sigma_{x\hat{x}}}{\sigma_x\sigma_{\hat{x}}}$$

where μ_x and $\mu_{\hat{x}}$ are the estimated means of the local variance, and σ_x and $\sigma_{\hat{x}}$ are the standard deviations of the local variance, respectively. $\sigma_{x\hat{x}}$ is the corresponding covariance.

The Structure Similarity Index Measure (SSIM) metric assesses the image quality from the perspective of structural information. The higher the value, the more similar two images are. The corresponding is given by [9]

$$\text{SSIM}(a, b) = \frac{(2\mu_a\mu_b + C_1)(2\sigma_{ab} + C_2)}{(\mu_a^2 + \mu_b^2 + C_1)(\sigma_a^2 + \sigma_b^2 + C_2)}, \quad (32)$$

where μ_a is the mean intensity of a ; μ_b is the mean intensity of b ; σ_a^2 is the variance of a ; and σ_b^2 is the variance of b . Constants C_1 and C_2 are included to avoid instabilities when the denominator is very close to 0. A higher SSIM indicates two images have more structural similarity, which means the stronger detail preservation in reconstruction.

4.1 Experiments on visual effect comparison

In Figs. 2, 3, and 4, we carry out the visual comparisons of the denoised MR images by the different denoising methods. It is observed that the proposed DoG-WNNM is very similar to the true image in visual perception, in which some edges and fine structures are better preserved, and it significantly improves the image sharpness. The reason is that our method sufficiently exploits the NSS structure of image and the DoG filter operation in each group patch procedure.

To further demonstrate the effectiveness of the DoG-WNNM, we compare the difference between the original image and denoising results in the third row of Figs. 2, 3, and 4. It is evident that all of denoising methods demonstrate remarkable performance in removing noise. The NLM can be effective to remove the noise, but it loses the parts of texture at edge regions, which enables blurring of the tissue structures of interest. By contrast, the images produced by DoG-WNNM exhibit less texture at the third row, and better preserve the edges and texture details. In Figs. 5 and 6, we further address the MR image denoising with the real data. Compared with the other competing algorithms, the output of the DoG-WNNM is visually more pleasant. Therefore, the proposed method enhances the edges and fine structures of images and effectively reduces the noise.

4.2 Experiments with different noise levels

The PSNR/RLNE and QILV results of all competing algorithms with different noise levels are shown in Fig. 7. The T1w/T2w lesions MR images with standard deviation σ from 6 to 72 are used in this experiments. It can be observed that all of PSNR/RLNE and QILV curves obtained by the proposed method outperform other compared methods.

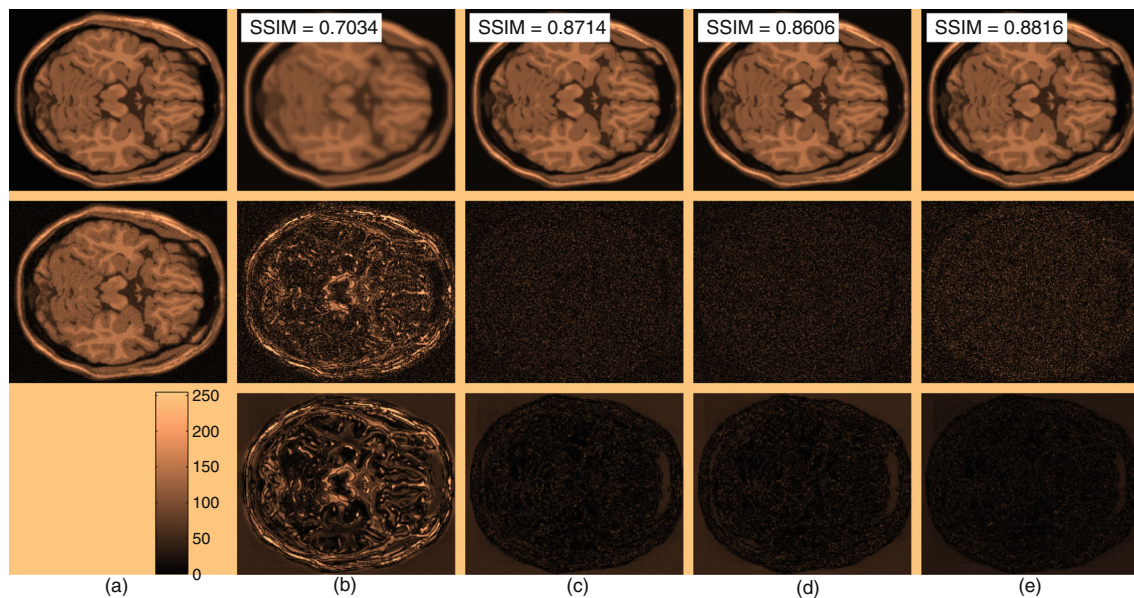


Fig. 2 MR image denoising for the proposed method and other competing methods on the T1w MR volume. **a** At the top is the original image and the middle is the image corrupted with a noisy image (standard deviation $\sigma = 7$). **b–e** At the top are the denoised images using

NLM, WNNM, WNNMD-N, and DoG-WNNM, respectively. In the middle are the corresponding residual images (the absolute difference between the denoised and noisy images), while in the bottom are the absolute differences between the original and denoised images

For quantitative analysis, the PSNR/RLNE and QILV values obtained by all competing algorithms with different standard deviation are reported in Table 1. With increasing noise level, the proposed method performs consistently better than other existing algorithms. In particular, the

average PSNR values of the DoG-WNNM are about 2.4 dB higher than NLM, 1.1 dB higher than WNNM, and 0.8 dB higher than WNNMD-N. The results reveal that the MR images with higher qualities are obtained by the proposed method in terms of PSNR/RLNE and QILV values.

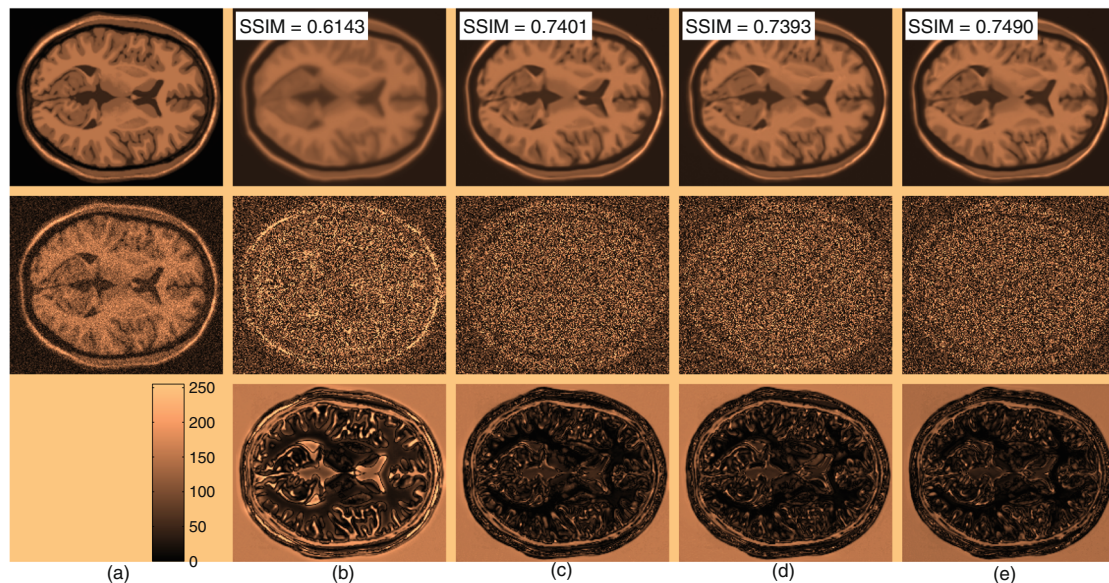


Fig. 3 MR image denoising for the proposed method and other competing methods on the T2w MR volume. **a** At the top is the original image and in the middle is the image corrupted with a noisy image (standard deviation $\sigma = 25$). **b–e** At the top are the denoised images

using NLM, WNNM, WNNMD-N, and DoG-WNNM, respectively. In the middle are the corresponding residual images (the absolute difference between the denoised and noisy images), while at the bottom are the absolute differences between the original and denoised images

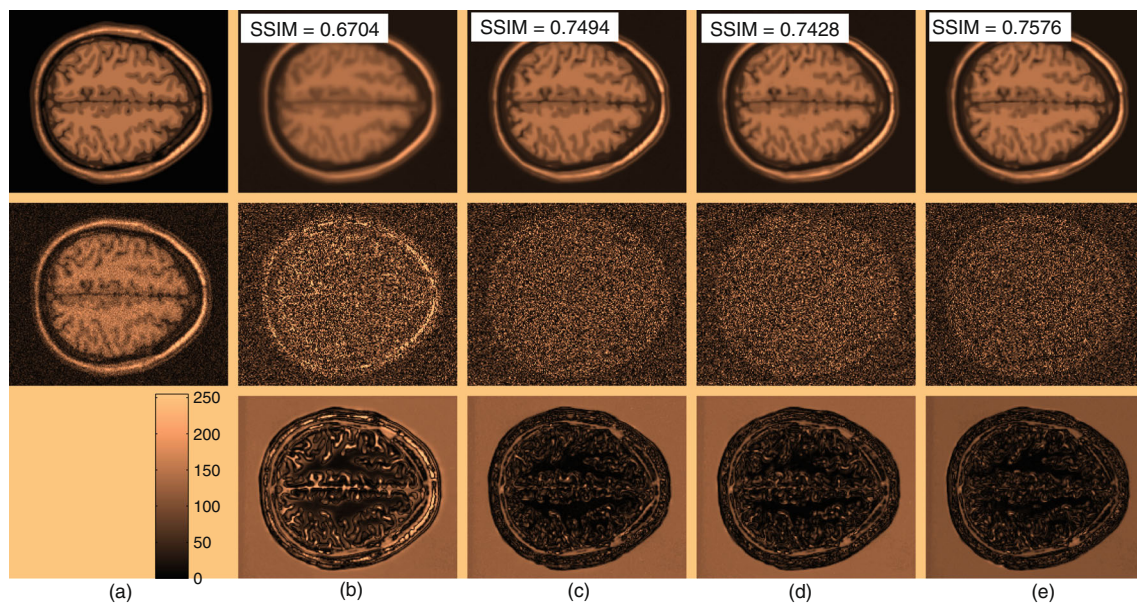


Fig. 4 MR image denoising for the proposed method and other competing methods on the T2w MR volume. **a** At the top is the original image and in the middle is the image corrupted with a noisy image (standard deviation $\sigma = 20$). **b–e** At the top are the denoised images

using NLM, WNNM, WNNMD-N, and DoG-WNNM, respectively. In the middle are the corresponding residual images (the absolute difference between the denoised and noisy images), while at the bottom are the absolute differences between the original and denoised images

4.3 The choice of patch size

Similar to nonlocal similarity methods, the patch size of the image is one of the key factors in denoising procedures. In general, a large size of image patch leads to smoothening in some fine-detailed regions. In contrast, a smaller size of the image patch increases the computational cost and also generates the noise artifacts.

To evaluate the influence of different patch sizes for the performance of the DoG-WNNM, we conduct a series of experiments that chooses the patch size from 4×4 to 13×13 . The PSNR, RLNE, and QILV indices obtained by the proposed method with standard deviation $\sigma = 15$ and $\sigma = 25$ are shown in Tables 2 and 3, respectively. From Tables 2 and 3, it can be observed that the denoised image with the 6×6 patch size can obtain more stable results in

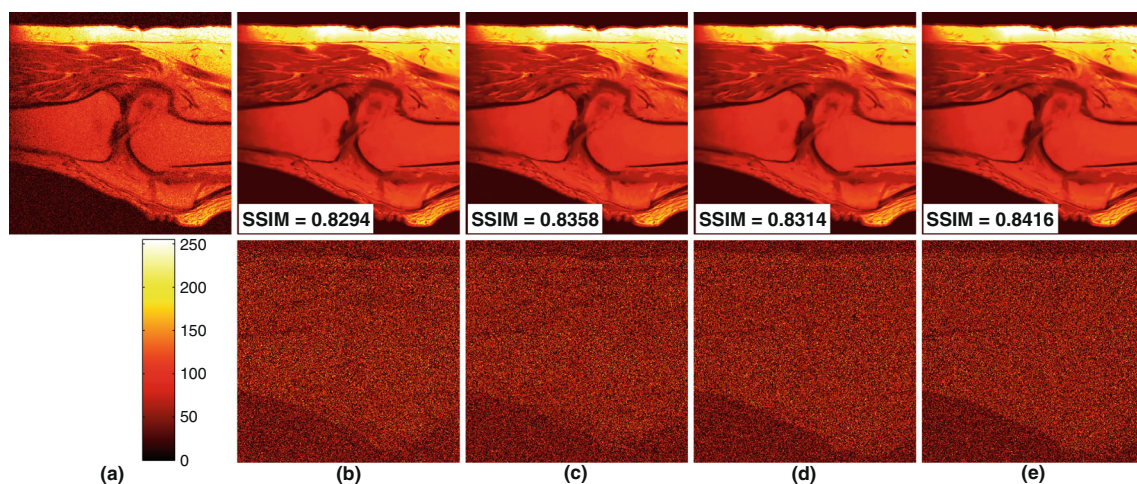


Fig. 5 MR image denoising for the proposed method and other competing methods. **a** At the top is the noisy MR image. **b–e** At the top are the denoised results obtained by NLM, WNNM, WNNMD-N, and

DoG-WNNM, respectively. At the bottom are the corresponding residual images (the absolute difference between the denoised and noisy images)

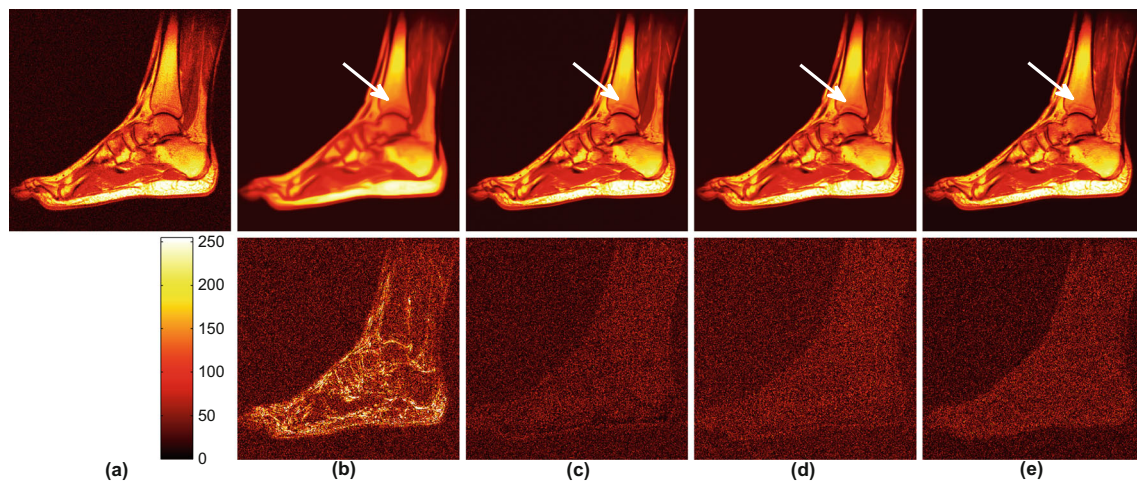


Fig. 6 MR image denoising for the proposed method and other competing methods. **a** At the top is the noisy MR image. **b–e** At the top are the denoised results obtained by NLM, WNNM, WNNMD-N, and

DoG-WNNM, respectively. At the bottom are the corresponding residual images (the absolute difference between the denoised and noisy images)

terms of the PSNR, RLNE, and QILV indices. Therefore, the image patch size of 6×6 is used in the proposed denoising method.

4.4 The comparison in time

We also provide computational time comparison for the proposed method and other competing methods. The running time is measured using MATLAB on a personal

computer with Intel Core i3 CPU, 8GB RAM, and Microsoft Windows 7 system. To be fair, the experiments were repeated 50 times and the average results are given in Fig. 8. The comparison shows that the proposed method obtains competitive denoising results, but it also increases the running time. The reason is the use of DoG filter and the noise variance estimation in the proposed method. However, total runtime overhead of the proposed method is quite small compared with the existing WNNMD-N.

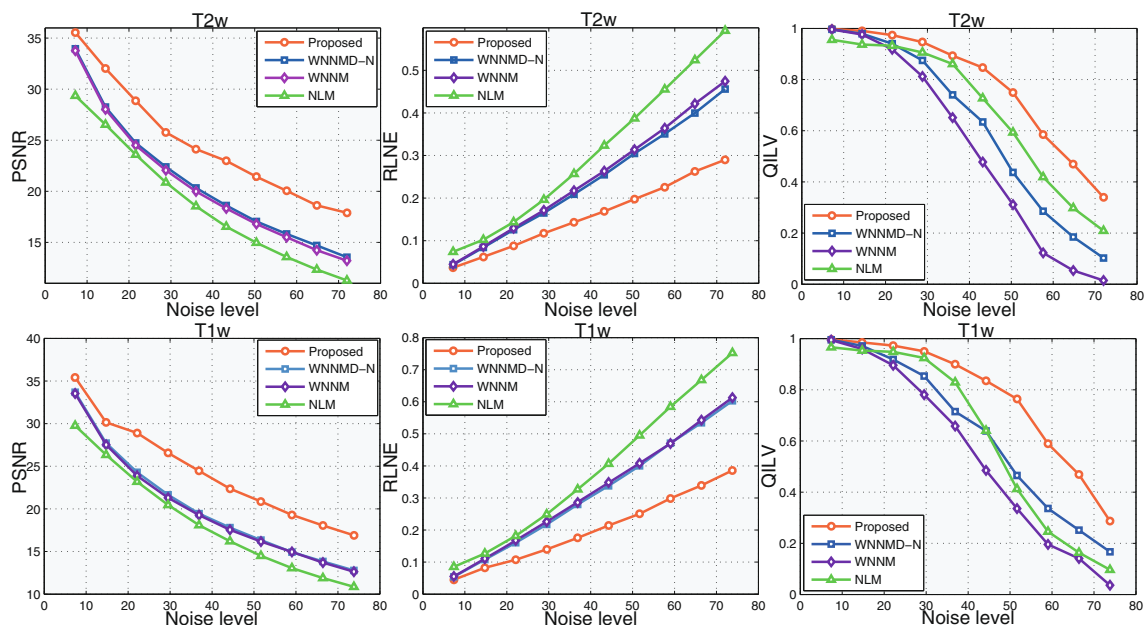


Fig. 7 PSNR (left), RLNE (middle), and QILV (right) of the compared methods for different volume types (T1w, T2w) and standard deviation from 6 to 72 with a step of 3

Table 1 The PSNR/RLNE/QILV/SSIM indices obtained by several image denoising methods

Methods	Indices	Noise deviation										
		$\sigma = 10$	$\sigma = 15$	$\sigma = 20$	$\sigma = 25$	$\sigma = 30$	$\sigma = 35$	$\sigma = 40$	$\sigma = 45$	$\sigma = 50$	$\sigma = 55$	$\sigma = 60$
NLM	PSNR	30.522	27.387	24.775	22.620	21.125	19.624	17.894	17.158	16.127	15.290	14.389
	RLNE	0.0836	0.0804	0.1621	0.2101	0.2595	0.2981	0.3512	0.3972	0.4481	0.4942	0.5489
	QILV	0.9739	0.9687	0.9533	0.9552	0.8658	0.8631	0.7332	0.7495	0.5915	0.5590	0.4849
	SSIM	0.8193	0.7903	0.7490	0.7088	0.6766	0.6289	0.5806	0.5421	0.5115	0.4686	0.4263
WNNM	PSNR	31.023	27.510	25.059	22.893	21.245	19.845	18.571	17.371	16.345	15.489	14.581
	RLNE	0.0814	0.0814	0.1618	0.2076	0.2510	0.2949	0.3415	0.3921	0.4413	0.4870	0.5407
	QILV	0.9932	0.9802	0.9655	0.9697	0.8819	0.8339	0.7348	0.7293	0.5994	0.5307	0.4938
	SSIM	0.8327	0.7939	0.7576	0.7174	0.6805	0.6360	0.6026	0.5488	0.5185	0.4747	0.4321
BM4D [35]	PSNR	30.535	26.344	24.476	21.993	21.011	19.445	17.874	16.713	16.193	13.768	13.058
	RLNE	0.0861	0.0907	0.1716	0.2297	0.2673	0.3084	0.3565	0.4027	0.4488	0.5499	0.4840
	QILV	0.9723	0.9746	0.9496	0.9210	0.8479	0.8228	0.7230	0.7141	0.5538	0.4924	0.4376
	SSIM	0.8197	0.7601	0.7399	0.6892	0.6730	0.6232	0.5800	0.5280	0.5137	0.4220	0.3870
PRI-NL-PCA [1]	PSNR	30.615	27.470	24.850	22.688	21.189	19.684	17.948	17.211	16.176	15.337	14.433
	RLNE	0.0838	0.0806	0.1625	0.2105	0.2600	0.2986	0.3519	0.3980	0.4490	0.4951	0.5499
	QILV	0.9769	0.9769	0.9562	0.9558	0.8685	0.8657	0.7354	0.7518	0.5933	0.5607	0.4863
	SSIM	0.8217	0.7927	0.7513	0.7110	0.6787	0.6308	0.5824	0.5438	0.5131	0.4701	0.4276
WNNMD-N	PSNR	30.925	27.748	25.102	22.918	21.404	19.883	18.130	17.385	16.340	15.492	14.579
	RLNE	0.0824	0.0793	0.1598	0.2070	0.2557	0.2937	0.3461	0.3914	0.4415	0.4869	0.5408
	QILV	0.9868	0.9815	0.9659	0.9678	0.8773	0.8745	0.7429	0.7594	0.5993	0.5664	0.4913
	SSIM	0.8301	0.8008	0.7589	0.7182	0.6856	0.6372	0.5883	0.5493	0.5183	0.4748	0.4320
DoG-WNNM	PSNR	32.074	28.846	26.295	23.859	22.439	20.958	19.595	19.191	17.612	17.458	16.253
	RLNE	0.0747	0.0694	0.1430	0.1991	0.2360	0.3055	0.3210	0.3792	0.3815	0.4259	0.4443
	QILV	0.9898	0.9847	0.9722	0.9734	0.8908	0.8648	0.7586	0.7580	0.6333	0.6127	0.5427
	SSIM	0.8609	0.8324	0.7950	0.7477	0.7187	0.6717	0.6358	0.6063	0.5587	0.5351	0.4816

Table 2 The PSNR, RLNE and QILV indices of denoised images with different patch size (Noise Level, $\sigma = 15$)

Patch size	PSNR	RLNE	QILV	SSIM
4×4	26.650	0.1442	0.9839	0.7153
6×6	26.211	0.1580	0.9855	0.7564
9×9	29.425	0.1201	0.9892	0.8896
11×11	27.236	0.1446	0.9821	0.8535
13×13	26.452	0.1677	0.9610	0.8473

Table 3 The PSNR, RLNE and QILV indices of denoised images with different patch size (Noise Level, $\sigma = 25$)

Patch size	PSNR	RLNE	QILV	SSIM
4×4	25.283	0.1809	0.8971	0.8204
6×6	26.096	0.1659	0.9415	0.8244
9×9	25.984	0.1767	0.9154	0.8242
11×11	22.577	0.2435	0.9076	0.6919
13×13	21.202	0.2915	0.7875	0.6283

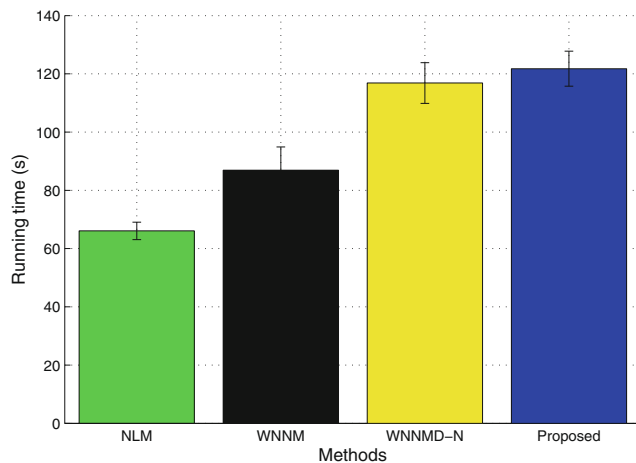


Fig. 8 Running time given by the proposed method and other competing methods ($\sigma=17$)

5 Conclusion

This paper proposes a novel denoising framework based on the DoG filter and nonlocal low-rank model, which improves the quality of denoised image. To demonstrate the effectiveness of the DoG-WNNM for MR image denoising, we combine the novel NSS prior of image and the extended DoG filter together to construct the low-rank regularization model. Moreover, the equivalent model is also proposed to solve the WNNM regularization model. Experimental results with different noise levels demonstrate the advantages of the DoG-WNNM. Compared with the existing denoising methods, the proposed method has better performance in preserving the edge structures and detailed textures of the image. Theoretical analysis about the Rician bias correction is planned as part of a future work.

Appendix

By the sparse representation of tight frame $D_l^H D_l = I$, let L_{Z_l} and L_{S_l} be the transform domain representation of Z_l and S_l , respectively, i.e., $L_{Z_l} = D_l Z_l$, $L_{S_l} = D_l S_l$. Then, the minimization problem of Eq. 12 can be given by:

$$\begin{aligned} & \min_{S_l} \frac{1}{2} \|Z_l - S_l\|_F^2 + \lambda \sum_{i=1}^r w_{l_i} \sigma_i(S_l) \\ &= \min_{S_l} \frac{1}{2} \|D_l Z_l - D_l S_l\|_F^2 + \lambda \sum_{i=1}^r w_{l_i} \sigma_i(L_{S_l}) \\ &\stackrel{(a)}{=} \min_{L_{S_l}} \frac{1}{2} \|L_{Z_l} - L_{S_l}\|_F^2 + \lambda \sum_{i=1}^r w_{l_i} \sigma_i(L_{S_l}) \end{aligned} \quad (33)$$

where (a) is from the tight frame and

$$\begin{aligned} \|Z_l - S_l\|_F^2 &= \text{tr}((Z_l - S_l) D_l^H D_l (Z_l - S_l)) \\ &= \|L_{Z_l} - L_{S_l}\|_F^2. \end{aligned}$$

Let the SVD of L_{Z_l} be $L_{Z_l} = U_l \Sigma_{Z_l} V_l^H$ and define the matrix $\Phi_l \triangleq U_l^H L_{S_l} V_l$, then the objective function in Eq. 33 can be rewritten as:

$$\begin{aligned} & \frac{1}{2} \|L_{Z_l} - L_{S_l}\|_F^2 + \lambda \sum_{i=1}^r w_{l_i} \sigma_i(\Phi_l) \\ &= \frac{1}{2} \|U_l (\Sigma_{Z_l} - \Phi_l) V_l^H\|_F^2 + \lambda \sum_{i=1}^r w_{l_i} \sigma_i(\Phi_l) \\ &= \frac{1}{2} \|\Sigma_{Z_l} - \Phi_l\|_F^2 + \lambda \sum_{i=1}^r w_{l_i} \sigma_i(\Phi_l) \\ &\stackrel{(a)}{=} \frac{1}{2} (\text{tr}(\Sigma_{Z_l}^2) + \text{tr}(\Sigma_{\Phi_l}^2)) + \text{tr}(\Sigma_{Z_l} U_{\Phi_l} \Sigma_{\Phi_l} V_{\Phi_l}^T) \\ &\quad + \lambda \sum_{i=1}^r w_{l_i} \sigma_i(\Phi_l) \\ &\stackrel{(b)}{\leq} \frac{1}{2} (\text{tr}(\Sigma_{Z_l}^2) + \text{tr}(\Sigma_{\Phi_l}^2)) + \sum_{i=1}^r \sigma_i(\Sigma_{Z_l}) \sigma_i(\Sigma_{\Phi_l}) \\ &\quad + \lambda \sum_{i=1}^r w_{l_i} \sigma_i(\Phi_l) \end{aligned} \quad (34)$$

where (a) is from the SVD of the Φ_l that $\Phi_l = U_{\Phi_l} \Sigma_{\Phi_l} V_{\Phi_l}^T$ and (b) follows from the fact that

$$\begin{aligned} \text{tr}(\Sigma_{Z_l} U_{\Phi_l} \Sigma_{\Phi_l} V_{\Phi_l}^T) &\leq \sum_{i=1}^r \sigma_i(\Sigma_{Z_l}) \sigma_i(U_{\Phi_l} \Sigma_{\Phi_l} V_{\Phi_l}^T) \\ &\leq \sum_{i=1}^r \sigma_i(\Sigma_{Z_l}) \sigma_i(\Sigma_{\Phi_l}) \end{aligned} \quad (35)$$

and $\sigma_i(\cdot)$ denotes the i -th largest singular value of a matrix.

Equation 35 holds if and only if $\Phi_l = \Sigma_{\Phi_l}$. Therefore, the rank minimization problem (33) is equivalent to the following minimization:

$$\min_{\Sigma_{\Phi_l}} \frac{1}{2} \|\Sigma_{Z_l} - \Sigma_{\Phi_l}\|_F^2 + \lambda \|W_l \Sigma_{\Phi_l}\|_* \quad (36)$$

where $W_l = \text{diag}\{w_{l_1}, w_{l_2}, \dots, w_{l_r}\}$ is a diagonal matrix.

Next, we will prove the minimization $\frac{1}{2} \|\Sigma_{Z_l} - \Sigma_{\Phi_l}\|_F^2 + \|W_l \Sigma_{\Phi_l}\|_*$ in Eq. 36 is equivalent to

$$\min_{\Sigma_{\Phi_l}} \frac{1}{2} \|W_l \Sigma_{Z_l} - W_l \Sigma_{\Phi_l}\|_F^2 + \lambda \|W_l \Sigma_{\Phi_l}\|_* \quad (37)$$

By the property of matrix norm, we have:

$$\begin{aligned} & \frac{1}{\|W_l^{-1}\|_F^2} \left(\|\Sigma_{Z_l} - \Sigma_{\Phi_l}\|_F^2 + \lambda \|W_l \Sigma_{\Phi_l}\|_* \right) \\ &\leq \frac{1}{\|W_l^{-1}\|_F^2} \left(\|W_l^{-1}\|_F^2 \|W_l (\Sigma_{Z_l} - \Sigma_{\Phi_l})\|_F^2 + \lambda \|W_l \Sigma_{\Phi_l}\|_* \right) \\ &= \|W_l (\Sigma_{Z_l} - \Sigma_{\Phi_l})\|_F^2 + \frac{\lambda}{\|W_l^{-1}\|_F^2} \|W_l \Sigma_{\Phi_l}\|_* \end{aligned} \quad (38)$$

Then, it easily follows that

$$\begin{aligned} & \|\Sigma_{Z_l} - \Sigma_{\Phi_l}\|_F^2 + \lambda \|W_l \Sigma_{\Phi_l}\|_* \\ &\leq \|W_l^{-1}\|_F^2 \|W_l (\Sigma_{Z_l} - \Sigma_{\Phi_l})\|_F^2 + \lambda \|W_l \Sigma_{\Phi_l}\|_* \end{aligned} \quad (39)$$

On the other hand,

$$\begin{aligned} & \frac{1}{\|W_l\|_F^2} \left(\|W_l(\Sigma_{z_l} - \Sigma_{\phi_l})\|_F^2 + \lambda \|W_l \Sigma_{\phi_l}\|_* \right) \\ & \leq \frac{1}{\|W_l\|_F^2} \left(\|W_l\|_F^2 \|\Sigma_{z_l} - \Sigma_{\phi_l}\|_F^2 + \lambda \|W_l \Sigma_{\phi_l}\|_* \right) \quad (40) \\ & = \|\Sigma_{z_l} - \Sigma_{\phi_l}\|_F^2 + \frac{\lambda}{\|W_l\|_F^2} \|W_l \Sigma_{\phi_l}\|_* \end{aligned}$$

It follows that

$$\begin{aligned} & \|W_l(\Sigma_{z_l} - \Sigma_{\phi_l})\|_F^2 + \lambda \|W_l \Sigma_{\phi_l}\|_* \\ & \leq \|W_l\|_F^2 \|\Sigma_{z_l} - \Sigma_{\phi_l}\|_F^2 + \lambda \|W_l \Sigma_{\phi_l}\|_* \quad (41) \end{aligned}$$

Suppose Ω_1 and Ω_2 are the solutions of the optimization problems (36) and (37), respectively. By Eq. 39, $\forall \Sigma'_{\phi_l} \in \Omega_1$, for any $\Sigma_{\phi_l} \in \Omega_2$, the following relationship can be easily obtained:

$$\begin{aligned} & \|\Sigma_{z_l} - \Sigma_{\phi_l}\|_F^2 + \lambda \|W_l \Sigma_{\phi_l}\|_* \\ & \leq \|\Sigma_{z_l} - \Sigma'_{\phi_l}\|_F^2 + \lambda \|W_l \Sigma'_{\phi_l}\|_* \quad (42) \\ & \leq \|W_l^{-1}\|_F^2 \|W_l(\Sigma_{z_l} - \Sigma'_{\phi_l})\|_F^2 + \lambda \|W_l \Sigma'_{\phi_l}\|_* \end{aligned}$$

which implies that $\Omega_2 \subseteq \Omega_1$.

According to Eq. 41, $\forall \Sigma_{\phi_l} \in \Omega_2$, for any $\Sigma'_{\phi_l} \in \Omega_1$, we can derive:

$$\begin{aligned} & \|W_l(\Sigma_{z_l} - \Sigma'_{\phi_l})\|_F^2 + \lambda \|W_l \Sigma'_{\phi_l}\|_* \\ & \leq \|W_l(\Sigma_{z_l} - \Sigma_{\phi_l})\|_F^2 + \lambda \|W_l \Sigma_{\phi_l}\|_* \quad (43) \\ & \leq \|W_l\|_F^2 \|\Sigma_{z_l} - \Sigma_{\phi_l}\|_F^2 + \lambda \|W_l \Sigma_{\phi_l}\|_* \end{aligned}$$

From the above result, we have $\Omega_1 \subseteq \Omega_2$. Furthermore, we easily obtain $\Omega_2 = \Omega_1$ which implies that the optimization problem (33) is equivalent to the minimization (37).

It completes the proof.

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